StreamPlus Premium-Tier Upgrade Targeting

Executive Summary

Recommendation: target subscribers only when predicted Premium-upgrade probability is at least 31%. On the holdout set, this rule improves estimated Net LTV impact from $3,561 at the default threshold to $5,147, while raising recall from 24.03% to 63.09%.

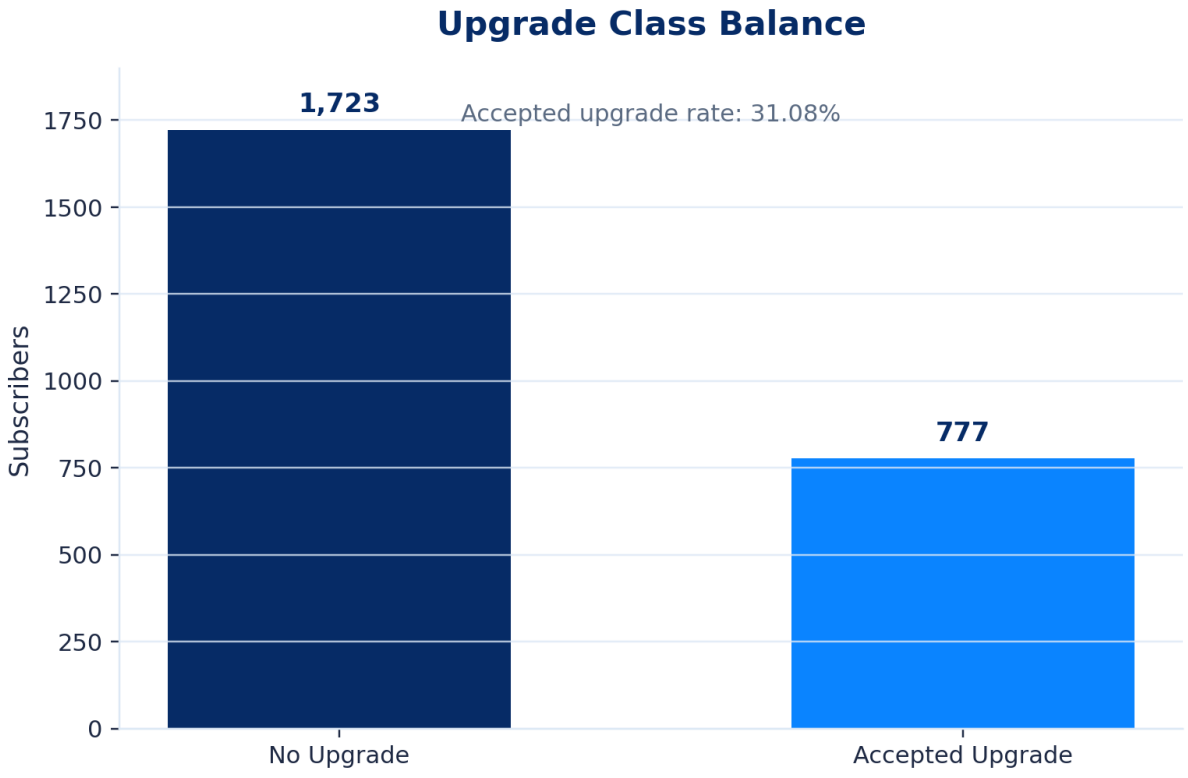
StreamPlus is a fictional direct-to-consumer streaming company seeking to grow average revenue per user by converting Basic subscribers to the Premium tier. Each successful upgrade is worth approximately $96 in incremental annual revenue, while each false-positive contact carries an expected $55 lifetime-value cost.

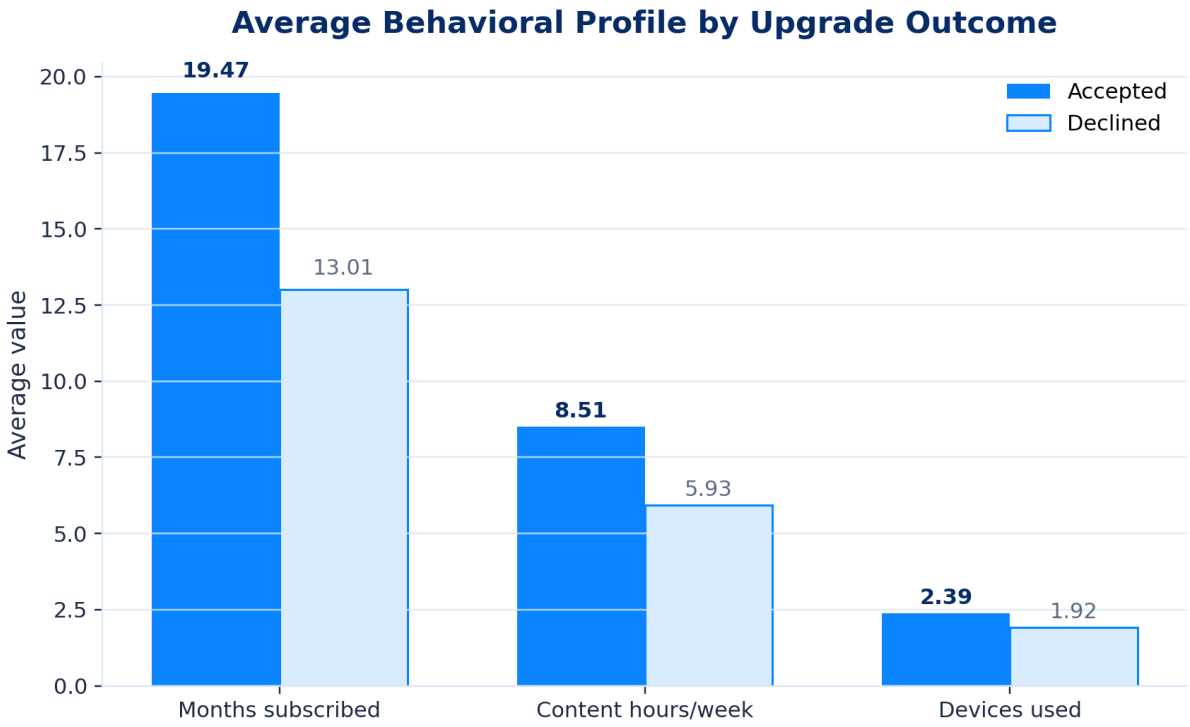
Business Problem

The core decision problem is not simply who is most likely to upgrade. It is which subscribers should be contacted so the campaign creates the highest expected business value.

Data and Workflow

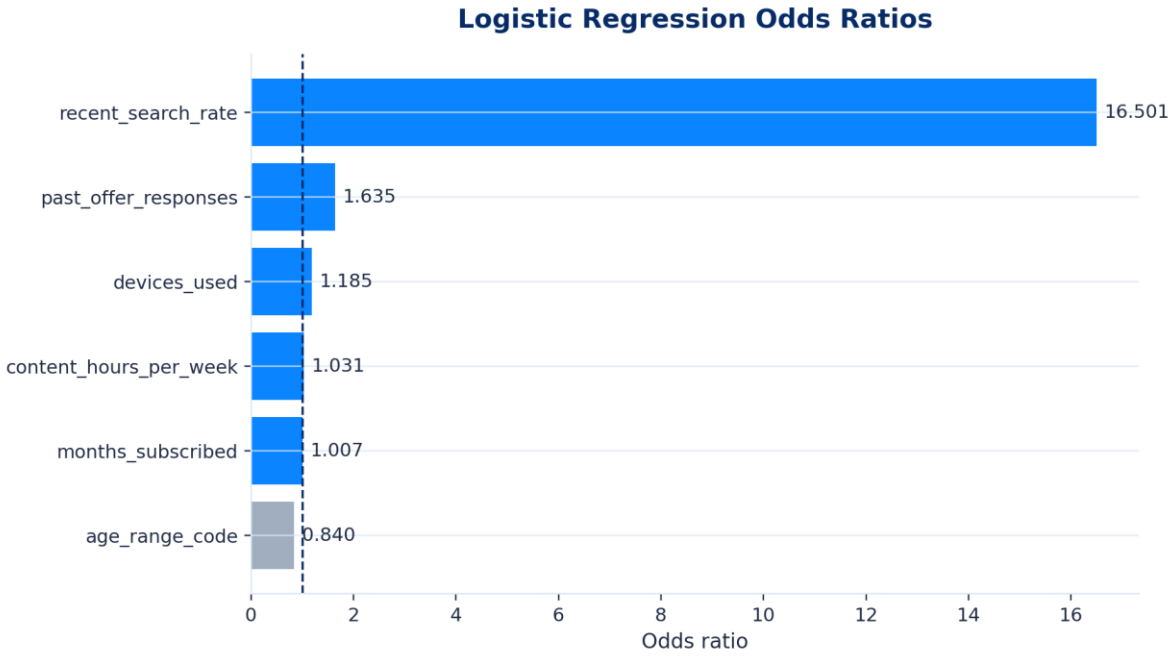
- 2,500 subscriber records
- No missing values
- 70/30 train-test split
- Logistic regression model
- Business threshold optimization based on Net LTV impact





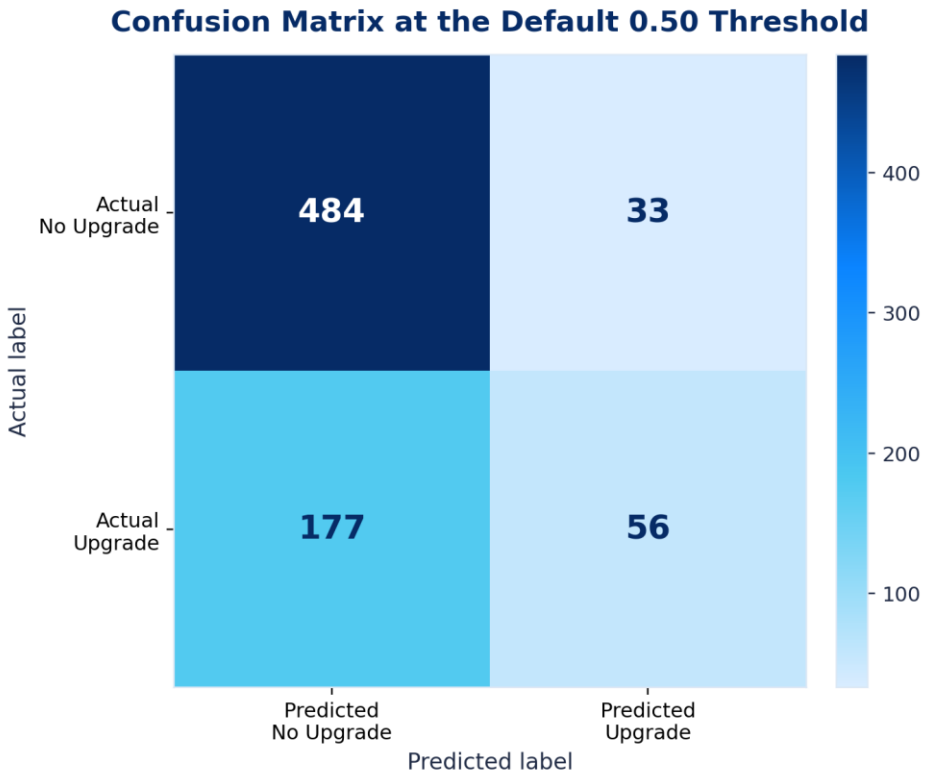
Key Findings

Accepted upgraders exhibited stronger behavioral engagement. Recent search activity, past offer responses, and multi-device usage emerged as the strongest positive signals.



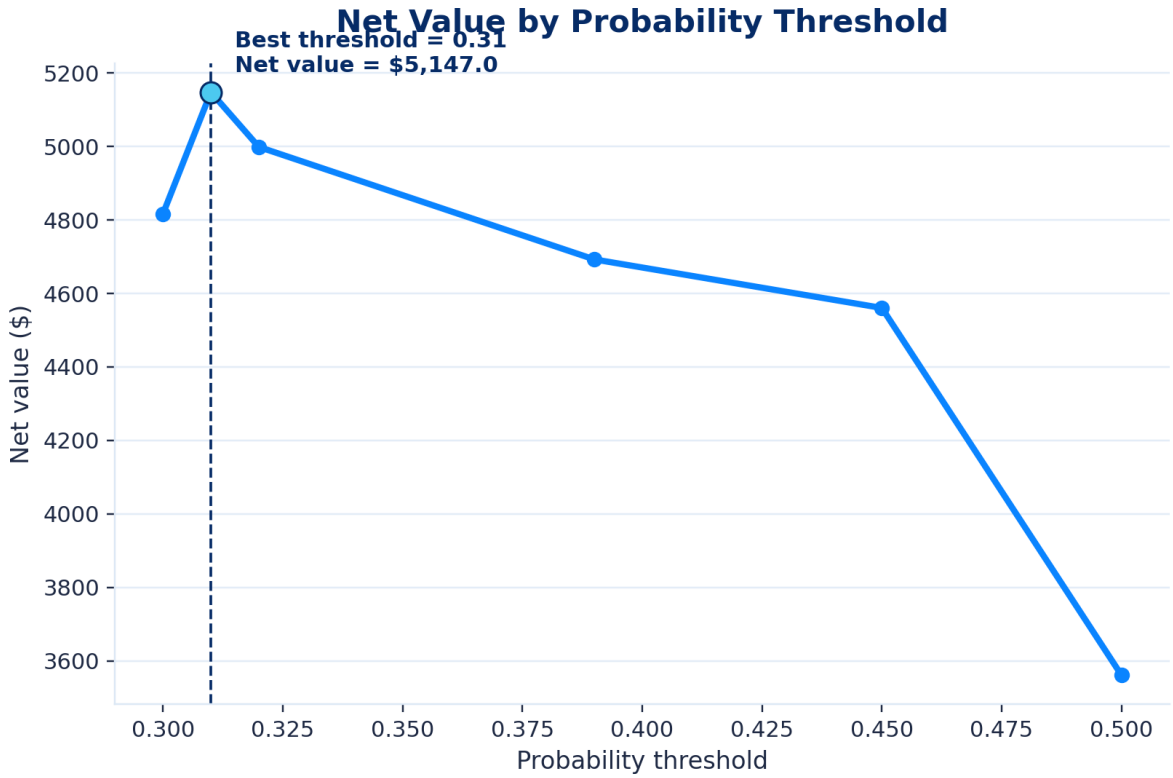
Baseline Performance

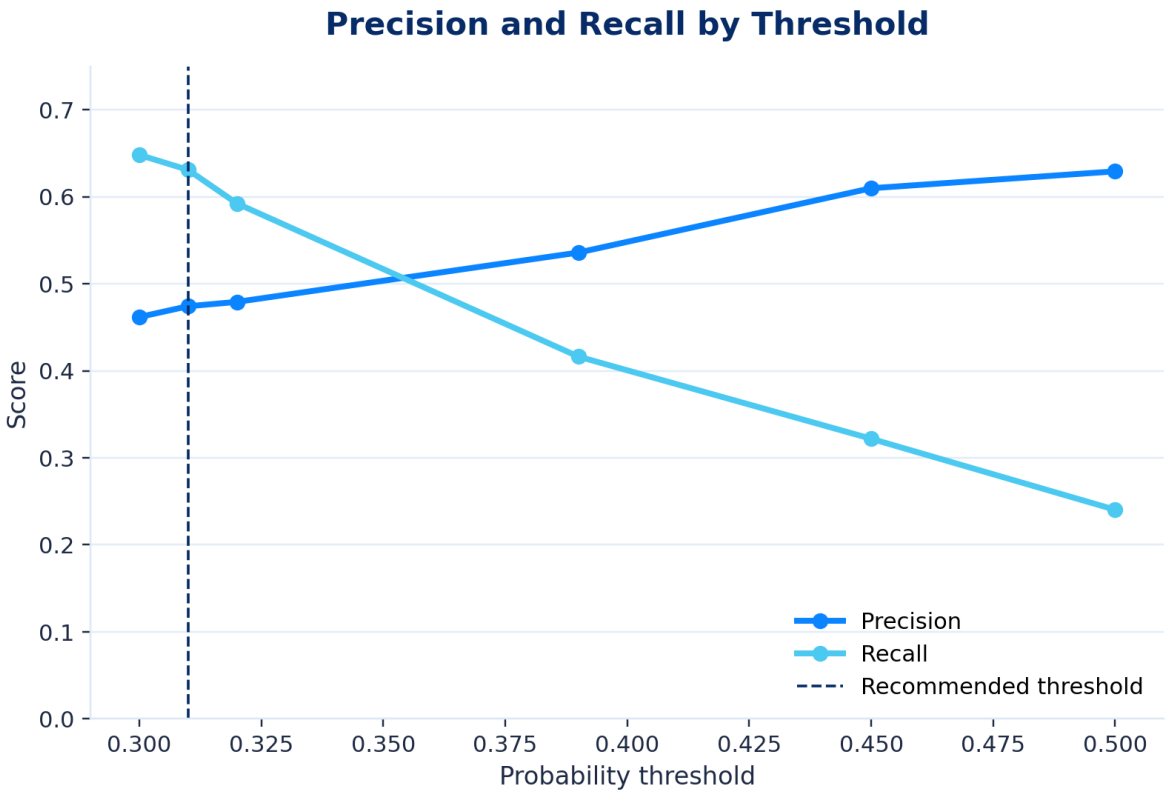
At the default 0.50 threshold, the model achieved 72.00% accuracy and 62.92% precision, but recall was only 24.03%, which means many real upgraders were missed.



Threshold Optimization

Using Net Value = (True Positives × 96) − (False Positives × 55), the best holdout-set threshold was 0.31.





Final Recommendation

Deploy the model as a targeting tool and send Premium offers only when predicted upgrade probability is at least 0.31. This recommendation is based on holdout-set performance and is therefore more defensible than optimizing directly on the full sample.

Limitations and Next Steps

The project uses a fictional business case, scenario-based economics, and a limited feature set. A real implementation should validate the economics through experimentation, expand the feature set, and compare additional models.